

Filip Bělík

filip.belik@utah.edu |  [filipbelik](#) |  [fbelik](#)

PhD Candidate

Department of Mathematics and Scientific Computing and Imaging Institute
University of Utah, Salt Lake City, USA

EDUCATION

• University of Utah

Mathematics PhD: Applied & Computational Mathematics

August 2022 - Present

Salt Lake City, UT, USA

- Co-advised by [Dr. Akil Narayan](#) and [Dr. Christel Hohenegger](#)
- Intended graduation in May 2027
- President of University's [SIAM Student chapter](#), Fall 2024-Spring 2026
- Organized [Wasatch SIAM Student Chapters Conference](#), Spring 2026
- Chair of Mathematics Graduate Student Recruitment Committee, Spring 2025
- GPA: 4.00

Coursework:

- MATH 5080 Statistical Inference I
- MATH 6010 Linear Models
- MATH 6210 Real Analysis
- MATH 6410 Ordinary Differential Equations
- MATH 6420 Partial Differential Equations
- MATH 6610 Analysis of Numerical Methods I
- MATH 6620 Analysis of Numerical Methods II
- MATH 6630 Numerical Method for Partial Differential Equations
- MATH 6710 Applied Linear Operators and Spectral Methods
- MATH 6720 Applied Complex Variables and Asymptotic Methods
- MATH 6740 Bifurcation Theory
- MATH 6750 Fluid Dynamics
- MATH 6880 Mathematics of Data Science
- MATH 7875 Advanced Optimization

• Gustavus Adolphus College

BA Honors Mathematics & BA Computer Science

September 2018 - May 2022

St. Peter, MN, USA

- Student host for [Nobel Conference 2021, Big Data](#)
- Mathematics, Computer Science, and Statistics (MCS) Department Assistant, 2021
- President of Club Tennis; Co-President of Coding Club; MCS Club; Running Club
- Cumulative GPA: 3.989
- Major GPA: 4.00

Coursework:

- MCS-150 Discrete Mathematics
- MCS-177 Computer Science I (Python)
- MCS-178 Computer Science II (Java/Kotlin/Assembly)
- MCS-220 Introduction to Analysis
- MCS-221 Linear Algebra
- MCS-222 Multivariable Calculus
- MCS-256 Discrete Calculus
- MCS-265 Theory of Computation
- MCS-270 Android Development
- MCS-284 Computer Organization (C)
- MCS-313/314 Modern Algebra I and II
- MCS-321 Theory of Complex Variables
- MCS-331 Real Analysis
- MCS-353 Continuous Dynamical Systems
- MCS-355 Scientific Computing
- MCS-357 Discrete Dynamical Systems
- MCS-375 Algorithms
- MCS-377 Networking

• East Ridge High School

Secondary Education

September 2015 - May 2018

Woodbury, MN, USA

- Varsity Tennis
- Mathematics Club
- Weighted GPA: 4.123
- Unweighted GPA: 3.814

PUBLICATIONS/PREPRINTS

1. Filip Bělík, Akil Narayan, and John Turnage. *Constructive Tchakaloff results and well-conditioned quadrature through randomized least squares*. 2026. arXiv: [2609.09506](https://arxiv.org/abs/2609.09506) [math.NA]. URL: <https://arxiv.org/abs/2609.09506>
2. Henry Crandall et al. "Cuffless hemodynamic monitoring with physics-informed machine learning models". en. In: *Nat. Commun.* (May 2026)
3. Filip Bělík, Yanlai Chen, and Akil Narayan. *Greedy Rational Approximation for Frequency-Domain Model Reduction of Parametric LTI Systems*. 2025. arXiv: [2512.23814](https://arxiv.org/abs/2512.23814) [math.NA]. URL: <https://arxiv.org/abs/2512.23814>
4. Filip Bělík, Jesse Chan, and Akil Narayan. *Efficient and Robust Carathéodory-Steinitz Pruning of Positive Discrete Measures*. 2025. arXiv: [2510.14916](https://arxiv.org/abs/2510.14916) [math.NA]. URL: <https://arxiv.org/abs/2510.14916>

TEACHING

- **Engineering Calculus II** Spring 2026
Instructor of Record University of Utah
- **Engineering Calculus I** Spring 2025
Instructor of Record University of Utah
- **Graduate Applied Linear Operators and Spectral Methods** Fall 2025
Teaching Assistant University of Utah
- **Graduate Numerical Linear Algebra** Spring 2024
Teaching Assistant University of Utah
- **Mathematics of Medicine** Spring 2023
Laboratory Instructor University of Utah
- **Mathematics of Medicine** Spring 2023
Laboratory Instructor University of Utah
- **Various courses in Mathematics, Computer Science, and Statistics Department** Spring 2019 - Spring 2022
Tutor, Teaching Assistant, and Grader Gustavus Adolphus College

WORK EXPERIENCE

- **University of Utah Mathematics Department** August 2022 - Present
Graduate Research and Teaching Assistant Salt Lake City, UT, USA
 - Instructor of record for MATH 1320, Engineering Calculus II, Spring 2026
 - Instructor of record for MATH 1310, Engineering Calculus I, Spring 2025
 - Research funding under Dr. Narayan, Dr. Christel Hohenegger, and a University of Utah funding incentive seed grant
 - Lab TA for MATH 4600 Mathematics in Medicine
- **Allianz Life** May 2021 - August 2021 & May 2026 - August 2026
Hedging Intern Golden Valley, MN, USA
 - Software development with C# and SQL
 - Implemented procedure for automating the labeling of incoming market data
 - WinApps development for sparse 3D data and assistance with software development life cycle
 - Study of mathematical finance and software implementation for delta and gamma hedging
- **Gustavus Mathematics and Computer Science Department** February 2019 - May 2022
TA & Tutor & Grader St. Peter, MN, USA
 - Computer Science I Teacher's Assistant, Grader, and Tutor (Python)
 - Computer Science II Teacher's Assistant and Tutoring (Kotlin and Java)
 - Discrete Mathematics Grader
 - Online volunteer tutoring during COVID semester

- **Her Next Play**

CRM Intern

June 2020 - August 2020

Remote

- Research and implementation of contact resource management software

- Presentation and instruction to executives

CONFERENCES

- **Wasatch SIAM Student Chapters Conference**

April 2026

University of Utah, Salt Lake City UT



- Talk: Frequency Domain Greedy Model Order Reduction for LTI Systems

- **SIAM Uncertainty Quantification**

March 2026

Hyatt Regency, Minneapolis MN

- **Joint Mathematics Meetings**

January 2026

Walter E. Washington Convention Center, Washington D.C.



- Talk: Greedy Rational Approximation for Model Reduction of LTI Systems

- **Wasatch SIAM Student Chapters Conference**

April 2025

Utah State University, Logan UT



- Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature

- **MAA North Central Section Meeting**

March 2025

St. Olaf College, Northfield MN



- Talk: Carathéodory-Steinitz Pruning for Numerical Quadrature

- **Model Reduction and Surrogate Modeling**

September 2024

Scripps Seaside Forum, La Jolla CA



- Talk: Greedy Frequency Domain Model Reduction for Parametric Systems: New Theory and Algorithms

- **NSF Computational Mathematics PI Meeting**

July 2024

University of Washington, Seattle WA



- Poster: Dynamic Bulk Conductivity in Radial Artery

- **Mathematical Opportunities in Digital Twins**

December 2023

George Mason University, Arlington VA



- Poster: Dynamic Bulk Conductivity in Radial Artery

- **Midstates Consortium for Math and Science**

November 2019

University of Chicago, Chicago IL



- Talk: The Propagation of Health-Related Habits on Twitter

RESEARCH

- **Model Order Reduction and Numerical Methods for Conservation Laws**

May 2023 - Present

with [Dr. Akil Narayan](#)

- Study and implementation of finite difference, finite volume, discontinuous Galerkin methods for conservation laws
- Study and implementation of parametric model order reduction methods for linear stationary and nonstationary parametric problems
- Development of open-source software package [ModelOrderReductionToolkit.jl](#)

- **Blood Flow Modeling for Conductivity and Uncertainty Quantification**

June 2022 - Present

with [Dr. Christel Hohenegger](#)

- Analytical modeling of arterial blood flow in arterial tree and resulting electrical properties
- Goal of better understanding relationship between blood pressure and blood-driven electrical properties at the wrist
- Work to understand impacts of physiological parameters through local and (Sobol-based) global sensitivity analyses
- Collaboration with [Henry Crandall](#) and [Dr. Benjamin Sanchez](#) (University of Utah Electrical Engineering) and Tyler Schuessler and [Dr. Braxton Osting](#) (University of Utah Mathematics)

- **A Constructive Proof for Tchakaloff Quadrature**

March 2025 - Present

with [Dr. Akil Narayan](#) and John Turnage

- Study of existence of stable quadrature rules for functionals over L^2
- Corresponding algorithm for construction of quadrature rules
- **Carathéodory-Steinitz Pruning** May 2023 - Present
with [Dr. Akil Narayan](#) and [Dr. Jesse Chan](#) (Rice University)
 - Algorithm design and stability analysis for constructing positive, efficient, nested quadrature rules
 - Testing of various methods for numerical quadrature and model order reduction applications
 - Development of open-source Julia package [CaratheodoryPruning.jl](#)
- **Uncertainty Quantification for Markov Decision Processes in Ecological Settings** May 2023 - Present
with [John Turnage](#), [Dr. Akil Narayan](#) and [Dr. Jody Reimer](#)
 - Study of use of decision-based models for ecological processes
 - Identify uncertainty quantification methods for Markov decision processes
- **Modeling Closed Vortices as Self-Avoiding Polygons** August 2021 - May 2022
with [Dr. Pavel Bělík](#) (Augsburg University) and [Dr. Thomas LoFaro](#) (Gustavus Adolphus College)
 - Undergraduate mathematics honors project
 - Modeled closed-loop vortices, such as dolphin bubble rings or smoke rings, as self-avoiding polygons in the cubic lattice
 - Implemented sets of transformations for use in Metropolis Markov Chain Monte-Carlo methods
- **Port-and-Sweep Solitaire Army Problem** May 2020 - September 2020
with [Dr. Jacob Siehler](#) (Gustavus Adolphus College)
 - Studied mathematics of [Port-and-Sweep Solitaire](#)
 - Used algebraic and computational techniques to solve one-dimensional army problem
- **The Propagation of Health-Related Habits on Twitter** May 2019 - November 2019
with [Dr. Louis Yu](#) and [Jeffery Engelhardt](#) (Gustavus Adolphus College)
 - Accepted as one of six first-year Gustavus students for ten-week research project under First-Year Research Experience (FYRE)
 - Use of various machine learning models in classification of tweets
 - Construction of listener to run over twelve-week period
 - Presented and attended research presentations at Midstates Consortium at University of Chicago

HONORS AND AWARDS

- **Outstanding Graduate Student** May 2025
University of Utah Mathematics Department
- **RTG: Optimization and Inversion Summer Grant** May 2023
University of Utah Mathematics Department 
- **Fulbright Canada Mitacs Globalink Research Internship** April 2021
Mitacs Globalink 
- **Hilding Research Fund** April 2020
Gustavus Adolphus College
- **First-Year Research Experience (FYRE)** April 2019
Gustavus Adolphus College 
- **Math Problem Solving Competition** 2018, 2019
Gustavus Adolphus College 
- **Dean's Scholarship** Fall 2018
Gustavus Adolphus College

SKILLS

- **Programming Languages:** Julia, Python, MATLAB, LaTeX, C, C++, Java, C#
- **Computer Operating Systems:** Windows, Linux, Max
- **Web Development:** HTML, JavaScript, CSS
- **Mathematical/Technical Tools:** Jupyter, LaTeX, Maple, SQL
- **Android App Development:** Kotlin
- **Competitive Programming:** ICPC, COMAP, Kattis (fbelik), Project Euler (fbelik)